Bloch

Malcolm

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Contents

Chapter 1

Construction of the Real Numbers

Definition 1.1. *Let S be a set. A* ***binary operation*** *on S is a function $S \times S \to S$. A* ***unary operation*** *on S is a function $S \to S$.*

Let S be a set and let $* : S \times S \to S$ be a binary operation. If $x, y \in S$, the correct way to write the result of the operation to the pair $\langle x, y \rangle$ would be $*(\langle x, y \rangle)$. However since this is a bit cumbersome one may write $x * y$ instead. Similarly if $\neg : S \to S$ is a unary operation and $x \in S$ one may choose to write $\neg x$ over $\neg(x)$.

1.1 Axioms for the Natural numbers

1.1.1 Peano Postulates

Axiom: Peano Postulates
There exists a set $\mathbb{N}$ with an element $1 \in \mathbb{N}$ and a function $s : \mathbb{N} \to \mathbb{N}$ that satisfy the following three properties.

- There is no $n \in \mathbb{N}$ such that $s(n) = 1$.
- The function s is injective.
- Let $G \subseteq \mathbb{N}$ be a set. Suppose that $1 \in G$, and that if $g \in G$ then $s(g) \in G$. Then $G = \mathbb{N}$.

It will be shown that the set $\mathbb{N}$ is unique. We can therefore give it a name.

Definition 1.2. *The set of* ***natural numbers*** *denoted $\mathbb{N}$, is the set the existence of which is given in the Peano Postulates.*

1.1.2 Consequences of Peano Postulates

Lemma 1.3. *Let $a \in \mathbb{N}$. Suppose that $a \neq 1$. Then there is a unique $b \in \mathbb{N}$ such that $a = s(b)$. (where existence of s and $\mathbb{N}$ are outlined by the peano axioms)*

Proof. First we prove existence. Let

$$G = \{1\} \cup \{c \in \mathbb{N} \mid \exists b \in \mathbb{N}(s(b) = c)\}$$

(which exists by the subset, pairing, and union axioms) We will prove that $G = \mathbb{N}$. To do this we need to show $1 \in G$, $G \subseteq \mathbb{N}$ and $\forall g \in G(s(g) \in G)$. The first two are straightforward. For the third suppose $g \in G$; we have two cases. First suppose $g \in \{1\}$; then $g = 1 \in \mathbb{N}$. Since $s : \mathbb{N} \to \mathbb{N}$, we then have $s(1) \in \mathbb{N}$. So by definition $s(1) \in \{c \in \mathbb{N} \mid \exists b \in \mathbb{N}(s(b) = c)\}$ and $s(g) = s(1) \in G$. Now suppose $g \in \{c \in \mathbb{N} \mid \exists b \in \mathbb{N}(s(b) = c)\}$; then by definition $g \in \mathbb{N}$ and $s(g) \in \mathbb{N}$ exists. By definition $s(g) \in \{c \in \mathbb{N} \mid \exists b \in \mathbb{N}(s(b) = c)\}$ so $s(g) \in G$. By the third peano postulate we then can assert that $G = \mathbb{N}$. So $a \in \mathbb{N} = G$. Since $a \neq 1$ we must then have $a \in \{c \in \mathbb{N} \mid \exists b \in \mathbb{N}(s(b) = c)\}$ and by definition, $\exists b \in \mathbb{N}(s(b) = a)$.

Now for uniqueness. Let m and n be arbitrary elements of $\mathbb{N}$ such that $a = s(m)$ and $a = s(n)$. Then $\langle m, a\rangle \in s$ and $\langle n, a\rangle \in s$. Since s is injective we must have $n = m$. □

Theorem 1.4. *Let H be a set, Let $e \in H$ and let $k : H \to H$ be a function. Then there is a unique function $f : \mathbb{N} \to H$ such that $f(1) = e$ and $f \circ s = k \circ f$.*

Proof. Letting H, e, and k be arbitrary such that $e \in H$ and $k : H \to H$. We will show existence and uniqueness separately. We start with uniqueness.

Let g and h be arbitrary such that $g : \mathbb{N} \to H$, $g(1) = e$, and $g \circ s = k \circ g$ and $h : \mathbb{N} \to H$, $h(1) = e$, and $h \circ s = k \circ h$. We will first define V as the unique set (guaranteed by the subset and extensionality axioms) such that

$$\forall a(a \in V \leftrightarrow a \in \mathbb{N} \wedge g(a) = h(a))$$

and show that $V = \mathbb{N}$. By peano's postulates, we need to show $V \subseteq \mathbb{N}$, $1 \in V$, and $\forall n \in V(s(n) \in V)$. The first is straightforward. For the second requirement, we know by peano's postulates that $1 \in \mathbb{N}$, so by our assumptions $g(1)$ and $h(1)$ exist and $g(1) = e = h(1)$. So $1 \in V$.

For the third requirement, let n be arbitrary and suppose that $n \in V$. Then by definition $n \in \mathbb{N}$ and $g(n) = h(n)$. Since $n \in \mathbb{N} = \mathrm{dom}(h \circ s)$, we have, by our assumptions,

$$h(s(n)) = h \circ s(n) = k \circ h(n) = k(h(n)) = k(g(n)) = k \circ g(n) = g \circ s(n) = g(s(n))$$

we know by definition $\mathrm{ran}(s) \subseteq \mathbb{N}$, so $s(n) \in \mathbb{N}$ and again by definition $s(n) \in V$. We can conclude that $V = \mathbb{N}$.

Suppose $\langle x, y\rangle \in g$, then $x \in \mathrm{dom} g = \mathbb{N} = V$. So by definition $y = g(x) = h(x)$, and $\langle x, y\rangle \in h$. Suppose $\langle x, y\rangle \in h$, then $x \in \mathrm{dom} h = \mathbb{N} = V$. So by definition $y = h(x) = g(x)$, and $\langle x, y\rangle \in g$.
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Now for existence. We need to come up with a function from $\mathbb{N}$ into H such that $f(1) = e$ and $f \circ s = k \circ f$. We first define C, where

$$\forall X(X \in C \leftrightarrow X \subseteq \mathbb{N} \times H \wedge \langle 1, e\rangle \in X \wedge \forall n \forall y(\langle n, y\rangle \in X \rightarrow \langle s(n), k(y)\rangle \in X))$$

(guaranteed by the subset and extensionality axioms.) We propose that $f = \bigcap C$ is the desired function. To ensure its existence, we need to show that C is nonempty. We will show that $\mathbb{N} \times H \in C$. For the first requirement, clearly $\mathbb{N} \times H \subseteq \mathbb{N} \times H$. Next, since $1 \in \mathbb{N}$ and $e \in H$ then $\langle 1, e\rangle \in \mathbb{N} \times H$. Finally, let n and y be arbitrary such that $\langle n, y\rangle \in \mathbb{N} \times H$. Since $s : \mathbb{N} \rightarrow \mathbb{N}$ and $k : H \rightarrow H$, we must then have $s(n) \in \mathbb{N}$ and $k(y) \in H$. So $\langle s(n), k(y)\rangle \in \mathbb{N} \times H$. $f = \bigcap C$ is therefore a valid candidate.

Before we prove the functionality of f, we first state a few facts about f. First we note that $f \in C$. Let arbitrary $x \in \bigcap C$. Since C is nonempty there exists $X \in C$, and by definition $x \in X \subseteq \mathbb{N} \times H$. So $\bigcap C \subseteq \mathbb{N} \times H$. Next let arbitrary $X \in C$. By definition $\langle 1, e\rangle \in X$, so $\langle 1, e\rangle \in \bigcap C$. Finally let n and y be arbitrary and suppose $\langle n, y\rangle \in \bigcap C$. We need to show that $\langle s(n), k(y)\rangle \in \bigcap C$. Let arbitrary $X \in C$. We have $\langle n, y\rangle \in X$. Since $X \in C$, by definition we must then have $\langle s(n), k(y)\rangle \in X$. We are done.

Next we note that $\forall X \in C(f \subseteq X)$. This is straightforward to prove. Finally we show

$$\begin{aligned}&\forall n \forall y([\langle n, y\rangle \in f \wedge \langle n, y\rangle \neq \langle 1, e\rangle] \\ &\qquad \rightarrow \exists m \exists u(\langle m, u\rangle \in f \wedge \langle n, y\rangle = \langle s(m), k(u)\rangle))\end{aligned}$$

Let n and y be arbitrary such that $\langle n, y\rangle \in f$ and $\langle n, y\rangle \neq \langle 1, e\rangle$. Suppose, for contradiction, that the statement after the implication is false. Then $\forall m \forall u(\langle m, u\rangle \in f \rightarrow \langle n, y\rangle \neq \langle s(m), k(u)\rangle)$. The set $f - \{\langle n, y\rangle\}$ exists by the subset axiom; we will show that this set is a member of C. We know that $f \in C$, so $f \subseteq \mathbb{N} \times H$, and it is straightforward to show that $(f - \{\langle n, y\rangle\}) \subseteq N \times H$. Since $\langle 1, e\rangle \neq \langle n, y\rangle$ and $\langle 1, e\rangle \in f$, we also have $\langle 1, e\rangle \in f - \{\langle n, y\rangle\}$. Finally, let arbitrary $\langle a, b\rangle \in f - \{\langle n, y\rangle\}$. Then $\langle a, b\rangle \in f \in C$ and $\langle s(a), k(b)\rangle \in f$. Since $\langle a, b\rangle \in f$, we must have, by our assumption, $\langle n, y\rangle \neq \langle s(a), k(b)\rangle$ and $\langle s(a), k(b)\rangle \in f - \{\langle n, y\rangle\}$. So $f - \{\langle n, y\rangle\} \in C$. It is straightforward to show that $f \nsubseteq f - \{\langle n, y\rangle\}$, contradicting the fact that $\forall X \in C(f \subseteq X)$.

We now prove functionality. We will do this by defining G:

$$\forall z(z \in G \leftrightarrow z \in \mathbb{N} \wedge \exists! x \in H(\langle z, x\rangle \in f))$$

(which exists by the subset axiom) and using the peano postulates to show that $G = \mathbb{N}$.
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We defined G:

$$\forall z(z \in G \leftrightarrow z \in \mathbb{N} \wedge \exists! x \in H(\langle z, x\rangle \in f))$$

To show that $G = \mathbb{N}$, we need to show that $G \subseteq \mathbb{N}$, $1 \in G$, and $\forall n \in G(s(n) \in G)$. The first requirement is straightforward. For the second, we already know by definition that $1 \in \mathbb{N}$. Since $f \in C$ we know by definition that $\langle 1, e\rangle \in f$, where $e \in H$. Let arbitrary $x' \in H$ such that $\langle 1, x'\rangle \in f$. Suppose for contradiction that $e \neq x'$. Then we have $\langle 1, e\rangle \neq \langle 1, x'\rangle$; so by the third statement proven earlier there exists m, u such that $\langle m, u\rangle \in f$ and $\langle 1, x'\rangle = \langle s(m), k(u)\rangle$. This means that $1 = s(m)$, where since $f \subseteq \mathbb{N} \times H$, $m \in \mathbb{N}$. This contradicts the peano postulates, which assert that $\neg\exists n \in \mathbb{N}(s(n) = 1)$.

For the third and final requirement, let n be an arbitrary element of G. Then by definition $n \in \mathbb{N}$ and there exists a unique x such that $\langle n, x\rangle \in f$. By definition we know $s(n) \in \mathbb{N}$, and since $f \in C$, we have $\langle s(n), k(x)\rangle \in f$, this verifies existence. Now let x' be arbitrary such that $\langle s(n), x'\rangle \in f$. By peano's postulates we know that $s(n) \neq 1$, so $\langle s(n), x'\rangle \neq \langle 1, e\rangle$. By the third statement proven earlier there exists m, u such that $\langle m, u\rangle \in f$ and $\langle s(n), x'\rangle = \langle s(m), k(u)\rangle$. So $s(n) = s(m)$ and $x' = k(u)$. Since s is injective we must have $m = n$. So $\langle n, u\rangle \in f$. By uniqueness of x we then have $u = x$. So $x' = k(u) = k(x)$, proving uniqueness. $s(n) \in G$.

We have shown that $G = \mathbb{N}$. It follows from this that $f : \mathbb{N} \to H$. For functionality, let $x \in \text{dom} f$. Then there exists y such that $\langle x, y\rangle \in f$. Since $f \subseteq \mathbb{N} \times H$ we have $x \in \mathbb{N} = G$, and by definition y is unique. Next to show $\text{dom} f = \mathbb{N}$, first let arbitrary $x \in \text{dom} f$, similarly it immediately follows that $x \in \mathbb{N}$. Now let $x \in \mathbb{N}$. Then $x \in G$ and by definition there exists $y \in H$ such that $\langle x, y\rangle \in f$, meaning $x \in \text{dom} f$. Finally to show $\text{ran} f \subseteq H$, let $y \in \text{ran} f$. Then there exists x such that $\langle x, y\rangle \in f \subseteq \mathbb{N} \times H$, so $y \in H$.

The fact that $\langle 1, e\rangle \in \bigcap C = f$ was already shown earlier. Finally we show $f \circ s = k \circ f$. Let $\langle x, y\rangle \in f \circ s$. Then $y = f \circ s(x) = f(s(x))$. Since $s(x) \neq 1$, $\langle s(x), y\rangle \neq \langle 1, e\rangle$ and as proven earlier there then exists m, u such that $\langle m, u\rangle \in f$ and $\langle s(x), y\rangle = \langle s(m), k(u)\rangle$. So $s(x) = s(m)$ and $y = k(u)$. Since s is injective $x = m$. We then have $\langle x, u\rangle \in f$ and $y = k(u) = k(f(x)) = k \circ f(x)$. so $\langle x, y\rangle \in k \circ f$. Next, suppose $\langle x, y\rangle \in k \circ f$. By definition there exists z such that $\langle x, z\rangle \in f$ and $\langle z, y\rangle \in k$. Since $f \in C$ we have $\langle s(x), k(z)\rangle \in f$. We then have $y = k(z) = f(s(x) = f \circ s(x)$ and $\langle x, y\rangle \in f \circ s$. We are done. $\square$

1.1.3 Addition on $\mathbb{N}$

Theorem 1.5. *There is a unique binary operation* $+ : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ *that satisfies the following two properties for all* $n, m \in \mathbb{N}$*:*

- $n + 1 = s(n)$
- $n + s(m) = s(n + m)$

Proof. We start with uniqueness. Let $+$ and $\oplus$ be arbitrary such that $+, \oplus : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$, $\forall m \in \mathbb{N} \forall n \in \mathbb{N}(+(\langle n, 1\rangle) = s(n) \wedge +(\langle n, s(m)\rangle) = s(+(\langle n, m\rangle)))$, and $\forall m \in \mathbb{N} \forall n \in \mathbb{N}(\oplus(\langle n, 1\rangle) = s(n) \wedge \oplus(\langle n, s(m)\rangle) = s(\oplus(\langle n, m\rangle)))$. We need to show $+ = \oplus$. We define G as

$$\forall x(x \in G \leftrightarrow x \in \mathbb{N} \wedge \forall n \in \mathbb{N}(+(\langle n, x\rangle) = \oplus(\langle n, x\rangle)))$$

(guaranteed by the subset axiom.) We will show that $G = \mathbb{N}$. To do this we need to show $G \subseteq \mathbb{N}$, $1 \in G$, and $\forall g \in G(s(g) \in G)$. The first requirement is straightforward. For the second requirement, we already know $1 \in \mathbb{N}$, and for arbitrary $n \in \mathbb{N}$, $+(\langle n, 1\rangle) = s(n) = \oplus(\langle n, 1\rangle)$.

For the third requirement, let arbitrary $g \in G$. Then by definition $g \in \mathbb{N}$ and $\forall n \in \mathbb{N}(+(\langle n, g\rangle) = \oplus(\langle n, g\rangle))$. Since $s : \mathbb{N} \to \mathbb{N}$ we know that $s(g) \in \mathbb{N}$ exists. Let n be arbitrary. By definition $+(\langle n, s(g)\rangle) = s(+(\langle n, g\rangle))$ and $\oplus(\langle n, s(g)\rangle) = s(\oplus(\langle n, g\rangle))$. By assumption we can assert that $+(\langle n, g\rangle) = \oplus(\langle n, g\rangle)$, so $+(\langle n, s(g)\rangle) = s(+(\langle n, g\rangle)) = s(\oplus(\langle n, g\rangle)) = \oplus(\langle n, s(g)\rangle)$. We conclude that $s(g) \in G$ and $G = \mathbb{N}$.

Let $\langle\langle x, y\rangle, z\rangle \in +$. By definition $y \in \mathbb{N} = G$ and $x \in \mathbb{N}$. So $+(\langle x, y\rangle) = \oplus(\langle x, y\rangle)$ and $z = \oplus(\langle x, y\rangle)$. So $\langle\langle x, y\rangle, z\rangle \in \oplus$. Now let $\langle\langle x, y\rangle, z\rangle \in \oplus$. A similar argument shows that $\langle\langle x, y\rangle, z\rangle \in +$.

Now for existence. We want to show

$$\exists + (+ : \mathbb{N} \times \mathbb{N} \to \mathbb{N} \wedge \forall m \in \mathbb{N} \forall n \in \mathbb{N}(+(\langle n, 1\rangle) = s(n) \\ \wedge +(\langle n, s(m)\rangle) = s(+(\langle n, m\rangle))))$$

Theorem 1.4 allows us to assert

$$\forall k \forall H \forall e([e \in H \wedge k : H \to H] \to \exists! f(f : \mathbb{N} \to H \wedge f(1) = e \wedge f \circ s = k \circ f))$$

It immediately follows that

$$\forall p \in \mathbb{N} \exists! f(f : \mathbb{N} \to \mathbb{N} \wedge f(1) = p \wedge f \circ s = s \circ f)$$

We will define $+$ as the set where

$$\forall z(z \in + \leftrightarrow \exists a \in \mathbb{N} \exists b \in \mathbb{N}(\exists f(f : \mathbb{N} \to \mathbb{N} \wedge f(1) = s(a) \wedge f \circ s = s \circ f \\ \wedge \exists c(\langle b, c\rangle \in f \wedge z = \langle\langle a, b\rangle, c\rangle))))$$

Existence follows from the subset axiom, where it can be shown that $+ \subseteq (\mathbb{N} \times \mathbb{N}) \times \mathbb{N}$.
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We defined $+$ as

$$\forall z(z \in + \leftrightarrow \exists a \in \mathbb{N} \exists b \in \mathbb{N}(\exists f(f : \mathbb{N} \to \mathbb{N} \wedge f(1) = s(a) \wedge f \circ s = s \circ f \\ \wedge \exists c(\langle b, c\rangle \in f \wedge z = \langle\langle a, b\rangle, c\rangle)))))$$

First we prove functionality. Let x be an arbitrary element of $\mathrm{dom}(+)$. Then there exists y such that $\langle x, y\rangle \in +$. Let y' be arbitrary such that $\langle x, y'\rangle \in +$. $\langle x, y\rangle \in +$ by definition means that there exists $a, b \in \mathbb{N}$, and $f : \mathbb{N} \to \mathbb{N}$ such that $f(1) = s(a)$ and $f \circ s = s \circ f$, and that there exists c such that $\langle b, c\rangle \in f$ and $\langle x, y\rangle = \langle\langle a, b\rangle, c\rangle$. $\langle x, y'\rangle \in +$ similarly implies the existence of $a_0, b_0 \in \mathbb{N}$ and $f_0 : \mathbb{N} \to \mathbb{N}$ such that $f_0(1) = s(a_0)$ and $f_0 \circ s = s \circ f_0$, and the existence of c_0 such that $\langle b_0, c_0\rangle \in f_0$ and $\langle x, y'\rangle = \langle\langle a_0, b_0\rangle, c_0\rangle$. Then $\langle a, b\rangle = x = \langle a_0, b_0\rangle$, so $a_0 = a$ and $b_0 = b$. It follows that $s(a_0) = s(a) \in \mathbb{N}$. This and

$$\forall p \in \mathbb{N} \exists! f(f : \mathbb{N} \to \mathbb{N} \wedge f(1) = p \wedge f \circ s = s \circ f)$$

implies that $f = f_0$. So we have $y = c = f(b) = f(b_0) = f_0(b_0) = c_0 = y'$.

Next we show that $\mathrm{dom}(+) = \mathbb{N} \times \mathbb{N}$. First let x be an arbitrary element of $\mathrm{dom}(+)$. Then there exists y such that $\langle x, y\rangle \in +$. By definition there then exists $a, b \in \mathbb{N}$ and there exists c such that $\langle x, y\rangle = \langle\langle a, b\rangle, c\rangle$. So $x = \langle a, b\rangle \in \mathbb{N} \times \mathbb{N}$.

Now suppose $x \in \mathbb{N} \times \mathbb{N}$. Then by definition there exists $a, b \in \mathbb{N}$ such that $x = \langle a, b\rangle$. $s(a) \in \mathbb{N}$ exists, so by theorem 1.4 there exists f such that $f : \mathbb{N} \to \mathbb{N}$, $f(1) = s(a)$, and $f \circ s = s \circ f$. Since $b \in \mathbb{N} = \mathrm{dom} f$, $f(b)$ exists and $\langle b, f(b)\rangle \in f$. Then by definition, $\langle\langle a, b\rangle, f(b)\rangle \in +$, so $x = \langle a, b\rangle \in \mathrm{dom}(+)$.

Next we show that $\mathrm{ran}(+) \subseteq \mathbb{N}$. Let arbitrary $y \in \mathrm{ran}(+)$. Then there exists x such that $\langle x, y\rangle \in +$. By definition there then exists $a, b \in \mathbb{N}$, $f : \mathbb{N} \to \mathbb{N}$, and c such that $\langle b, c\rangle \in f$ and $\langle x, y\rangle = \langle\langle a, b\rangle, c\rangle$. So we have $y = c = f(b) \in \mathbb{N}$.

Now we verify the two properties of addition, that is,

$$\forall m \in \mathbb{N} \forall n \in \mathbb{N}(+(\langle n, 1\rangle) = s(n) \wedge +(\langle n, s(m)\rangle) = s(+(\langle n, m\rangle)))$$

Let arbitrary $n, m \in \mathbb{N}$. We know that $+ : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$, so $\langle n, 1\rangle \in \mathbb{N} \times \mathbb{N} = \mathrm{dom}(+)$ and there exists y such that $\langle\langle n, 1\rangle, y\rangle \in +$. By definition there then exists $a, b \in \mathbb{N}$ and $f : \mathbb{N} \to \mathbb{N}$ such that $f(1) = s(a)$, and there exists c such that $\langle b, c\rangle \in f$ and $\langle\langle n, 1\rangle, y\rangle = \langle\langle a, b\rangle, c\rangle$. Then $n = a$, $1 = b$, and $y = c$. We then have $\langle 1, y\rangle \in f$ and $y = f(1) = s(a) = s(n)$. We conclude that $\langle\langle n, 1\rangle, s(n)\rangle \in +$ and $+(\langle n, 1\rangle) = s(n)$.
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Now for the second property. We know that $s(m) \in \mathbb{N}$. So $\langle n, s(m)\rangle \in \mathbb{N} \times \mathbb{N} = \text{dom}(+)$ and there exists y such that $\langle\langle n, s(m)\rangle, y\rangle \in +$. By definition there then exists $a, b \in \mathbb{N}$ such that $f : \mathbb{N} \to \mathbb{N}$, $f(1) = s(a)$, and $f \circ s = s \circ f$, and there exists c such that $\langle b, c\rangle \in f$ and $\langle\langle n, s(m)\rangle, y\rangle = \langle\langle a, b\rangle, c\rangle$. We then have $n = a$, $s(m) = b$, and $y = c = f(b) = f(s(m)) = s(f(m))$. We know that $\langle n, m\rangle \in \mathbb{N} \times \mathbb{N} = \text{dom}(+)$, so there exists y_0 such that $\langle\langle n, m\rangle, y_0\rangle \in +$. Again by definition there exists $a_0, b_0 \in \mathbb{N}$ and $f_0 : \mathbb{N} \to \mathbb{N}$ such that $f_0(1) = s(a_0)$ and $f_0 \circ s = s \circ f_0$ and there exists c_0 such that $\langle b_0, c_0\rangle \in f_0$ and $\langle\langle n, m\rangle, y_0\rangle = \langle\langle a_0, b_0\rangle, c_0\rangle$. Since it follows that $a_0 = n = a$ and $s(a_0) = s(a)$, theorem 1.4 allows us to assert $f = f_0$. Putting it all together, we have $+(\langle n, m\rangle) = y_0 = c_0 = f_0(b_0) = f_0(m) = f(m)$. We conclude that $+(\langle n, s(m)\rangle) = y = s(f(m)) = s(+(\langle n, m\rangle))$. $\square$

1.1.4 Multiplication on $\mathbb{N}$

Theorem 1.6. *There is a unique binary operation* $\cdot : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ *that satisfies the following two properties for all* $n, m \in \mathbb{N}$*:*

- $n \cdot 1 = n$
- $n \cdot s(m) = (n \cdot m) + n$

Proof. We start with uniqueness. Let $\cdot, \odot$ be arbitrary. Suppose $\cdot : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ and $\forall m \in \mathbb{N} \forall n \in \mathbb{N}(\cdot(\langle n, 1\rangle) = n \wedge \cdot(\langle n, s(m)\rangle) = +(\langle \cdot(\langle n, m\rangle), n\rangle))$. Similarly suppose $\odot : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ and $\forall m \in \mathbb{N} \forall n \in \mathbb{N}(\odot(\langle n, 1\rangle) = n \wedge \odot(\langle n, s(m)\rangle) = +(\langle \odot(\langle n, m\rangle), n\rangle))$. We want to show that $\cdot = \odot$. We do this by defining G such that

$$\forall z(z \in G \leftrightarrow z \in \mathbb{N} \wedge \forall n \in \mathbb{N}(\cdot(\langle n, z\rangle) = \odot(\langle n, z\rangle)))$$

(where existence is assured by the subset axiom) and showing that $G = \mathbb{N}$.

We need to show $G \subseteq \mathbb{N}$, $1 \in G$, and $\forall g \in G(s(g) \in G)$. The first requirement is straightforward. For the second, we already know $1 \in \mathbb{N}$, and see that for arbitrary $n \in \mathbb{N}$ we have $\cdot(\langle n, 1\rangle) = n = \odot(\langle n, 1\rangle)$.

For the third requirement let arbitrary $g \in G$. Then by definition $g \in \mathbb{N}$ and $\forall n \in \mathbb{N}(\cdot(\langle n, g\rangle) = \odot(\langle n, g\rangle))$. We know that $s(g) \in \mathbb{N}$ exists. So since $\langle n, s(g)\rangle \in \mathbb{N} \times \mathbb{N} = \text{dom}(\cdot)$, we have, by our assumptions, $\cdot(\langle n, s(g)\rangle) = +(\langle \cdot(\langle n, g\rangle), n\rangle) = +(\langle \odot(\langle n, g\rangle), n\rangle) = \odot(\langle n, s(g)\rangle)$. So $s(g) \in G$ and $G = \mathbb{N}$.

To prove the desired result, let arbitrary $z \in \cdot \subseteq (\mathbb{N} \times \mathbb{N}) \times \mathbb{N}$. Then there exists $a_0, a_1, b \in \mathbb{N}$ such that $\langle\langle a_0, a_1\rangle, b\rangle = z \in \cdot$. So $b = \cdot(\langle a_0, a_1\rangle)$. Since $a_1 \in \mathbb{N} = G$, we can then conclude that $b = \cdot(\langle a_0, a_1\rangle) = \odot(\langle a_0, a_1\rangle)$. So $z = \langle\langle a_0, a_1\rangle, b\rangle \in \odot$. Next suppose $z \in \odot$. It then follows from a similar argument that $z \in \cdot$. We are done.
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Now for existence. We start by defining a function h for all q, where

$$\forall q \exists h \forall z (z \in h \leftrightarrow \exists m \in \mathbb{N} \exists x \in \mathbb{N}(z = \langle m, x\rangle \wedge \langle\langle m, q\rangle, x\rangle \in +))$$

(which exists by the subset axiom, where $\mathbb{N} \times \mathbb{N}$ is a bounding set. Uniqueness for each q follows from extensionality.) We show that for all $q \in \mathbb{N}$, this h is a function from $\mathbb{N}$ into $\mathbb{N}$. That is

$$\forall q \in \mathbb{N}[\forall z (z \in h \leftrightarrow \exists m \in \mathbb{N} \exists x \in \mathbb{N}(z = \langle m, x\rangle \wedge \langle\langle m, q\rangle, x\rangle \in +)) \rightarrow h : \mathbb{N} \rightarrow \mathbb{N}]$$

Let arbitrary $q \in \mathbb{N}$. Suppose that the statement before the implication is true. First to prove functionality. Let $a \in \mathrm{dom} h$. Then there exists b such that $\langle a, b\rangle \in h$. Let b' be arbitrary such that $\langle a, b'\rangle \in h$. By our assumptions we must have $a, b, b' \in \mathbb{N}$, $+(\langle a, q\rangle) = b$, and $+(\langle a, q\rangle) = b'$. Since $+ : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ we must have $b = b'$. Next for the domain bound suppose arbitrary $a \in \mathrm{dom} h$. Then there exists b such that $\langle a, b\rangle \in h$. It immediately follows that $a \in \mathbb{N}$. Suppose instead that $a \in \mathbb{N}$. Then $\langle a, q\rangle \in \mathbb{N} \times \mathbb{N} = \mathrm{dom}(+)$. So $+(\langle a, q\rangle) \in \mathbb{N}$ exists and by definition, $\langle a, +(\langle a, q\rangle)\rangle \in h$. So $a \in \mathrm{dom} h$. Finally for the range bound, let arbitrary $b \in \mathrm{ran} h$. Then there exists a such that $\langle a, b\rangle \in h$. It immediately follows by definition that $b \in \mathbb{N}$. We have therefore shown

$$\begin{gathered}\forall q \in \mathbb{N} \exists! h (\forall z (z \in h \leftrightarrow \exists m \in \mathbb{N} \exists x \in \mathbb{N}(z = \langle m, x\rangle \wedge \langle\langle m, q\rangle, x\rangle \in +)) \\ \wedge\, h : \mathbb{N} \rightarrow \mathbb{N})\end{gathered}$$

Theorem 1.4 states

$$\forall H \forall q \forall h ([q \in H \wedge h : H \rightarrow H] \rightarrow \exists! g (g : \mathbb{N} \rightarrow H \wedge g(1) = q \wedge g \circ s = h \circ g))$$

which allows us to assert

$$\forall q \forall h ([q \in \mathbb{N} \wedge h : \mathbb{N} \rightarrow \mathbb{N}] \rightarrow \exists! g (g : \mathbb{N} \rightarrow \mathbb{N} \wedge g(1) = q \wedge g \circ s = h \circ g))$$

This, combined with the result we just proved, allows us to state

$$\begin{aligned}\forall q \in \mathbb{N} \exists! h (\forall z (z \in h \leftrightarrow \exists m \in \mathbb{N} \exists x \in \mathbb{N}(z = \langle m, x\rangle \wedge \langle\langle m, q\rangle, x\rangle \in +)) \\ \wedge\, h : \mathbb{N} \rightarrow \mathbb{N} \wedge \exists! g (g : \mathbb{N} \rightarrow \mathbb{N} \wedge g(1) = q \wedge g \circ s = h \circ g))\end{aligned}$$

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We showed

$$\forall q \in \ltimes \exists! h(\forall z(z \in h \leftrightarrow \exists m \in \ltimes \exists x \in \ltimes (z = \langle m, x\rangle \wedge \langle\langle m, q\rangle, x\rangle \in +))$$
$$\wedge\, h : \ltimes \to \ltimes \wedge \exists! g(g : \ltimes \to \ltimes \wedge g(1) = q \wedge g \circ s = h \circ g))$$

Now we construct $\cdot$ as follows

$\square$